10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 3.2 — Types of Matrices

Unit 3 | Solved Exercise | New Book (PECTAA)

Exercise 3.2 (Solved)

1. From the following matrices identify unit matrices, row matrices, column matrices and null matrices.

Row matrix: only one row. Column matrix: only one column. Unit (identity) matrix: square matrix whose diagonal elements are 1 and all other elements are 0. Null matrix: every element is 0.

$$A = \begin{bmatrix} 5 & 7 & 8 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$E = \begin{bmatrix} 6 \\ 0 \\ 8 \end{bmatrix}$$

$$F = \begin{bmatrix} 7 \\ 1 \\ 9 \end{bmatrix}$$

Identification

A = [5 7 8] has one row only

∴ A is a row matrix

B = [0] has one row and one column, and its only element is 0

∴ B is a null matrix (as well as a row / column matrix of order 1 × 1)

C has 1 on the diagonal and 0 elsewhere

∴ C is a unit matrix

D has every element equal to zero

∴ D is a null matrix

E and F have only one column each

∴ E and F are column matrices

Answer: Row: A Column: E, F Unit: C Null: B, D

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2. Identify the type of the given matrices as row, column, square and rectangular matrices.

Square matrix: number of rows = number of columns. Rectangular matrix: number of rows ≠ number of columns.

$$A = \begin{bmatrix} 5 \\ 3 \\ 4 \end{bmatrix} \quad (3 \times 1)$$

$$B = \begin{bmatrix} 1 & 6 \\ 4 & 1 \end{bmatrix} \quad (2 \times 2)$$

$$C = \begin{bmatrix} 3 & 5 & 8 \\ 0 & 4 & -2 \end{bmatrix} \quad (2 \times 3)$$

$$D = \begin{bmatrix} 5 & -5 \\ 2 & 7 \end{bmatrix} \quad (2 \times 2)$$

$$E = \begin{bmatrix} 3 & 2 \\ 4 & 1 \\ 5 & 0 \end{bmatrix} \quad (3 \times 2)$$

$$F = \begin{bmatrix} 5 & -3 & 7 \end{bmatrix} \quad (1 \times 3)$$

$$G = \begin{bmatrix} 3 & 0 & 1 \\ 1 & 3 & 4 \\ 5 & 2 & -3 \end{bmatrix} \quad (3 \times 3)$$

$$H = \begin{bmatrix} 3 & 5 \\ 4 & 4 \\ 5 & 2 \end{bmatrix} \quad (3 \times 2)$$

Identification

A is 3×1, so it has one column only $\Longrightarrow$ column matrix

B and D are 2×2 $\Longrightarrow$ square matrices

C is 2×3 $\Longrightarrow$ rectangular matrix

E is 3×2 $\Longrightarrow$ rectangular matrix

F is 1×3 $\Longrightarrow$ row matrix

G is 3×3 $\Longrightarrow$ square matrix

H is 3×2 $\Longrightarrow$ rectangular matrix

Answer: Column: A Row: F Square: B, D, G Rectangular: C, E, H

3. Identify diagonal, scalar and unit matrices.

Diagonal matrix: square matrix whose non-diagonal elements are all zero. Scalar matrix: a diagonal matrix whose diagonal elements are all equal. Unit matrix: a scalar matrix whose diagonal elements are all 1.

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$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

$$C = \begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix}$$

$$D = \begin{bmatrix} 7 & 0 \\ 0 & 6 \end{bmatrix}$$

$$E = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

Identification

In every matrix above the non-diagonal elements are zero

$\therefore$ A, B, C, D, E are all diagonal matrices

In B the diagonal elements are equal (5 and 5) $\Longrightarrow$ B is a scalar matrix

In E the diagonal elements are equal (2 and 2) $\Longrightarrow$ E is a scalar matrix

In A the diagonal elements are 1 and 1 $\Longrightarrow$ A is a unit matrix

In C and D the diagonal elements are not equal $\Longrightarrow$ only diagonal matrices

Answer: Diagonal: A, B, C, D, E Scalar: B, E Unit: A

4. Find the transpose of each of the following matrices:

Transpose: interchange rows and columns. If A is of order m × n, then A^t is of order n × m.

(i)

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$$

$$A^{t} = \begin{bmatrix} 2 & 4 \\ 3 & 5 \end{bmatrix}$$

Answer: A^t = 2, 4/3, 5 as shown above

(ii)

$$B = \begin{bmatrix} 3 \\ 7 \\ 6 \end{bmatrix}$$

$$B^{t} = \begin{bmatrix} 3 & 7 & 6 \end{bmatrix}$$

Answer: B^t = [3 7 6] (order 1 × 3)

(iii)

$$C = \begin{bmatrix} 5 & -2 & 4 \end{bmatrix}$$

$$C^{t} = \begin{bmatrix} 5 \\ -2 \\ 4 \end{bmatrix}$$

Answer: C^{t} = column [5 −2 4]ᵗ (order 3 × 1)

(iv)

$$D = \begin{bmatrix} 2 & 5 \\ 3 & 6 \\ 4 & 7 \end{bmatrix}$$

$$D^{t} = \begin{bmatrix} 2 & 3 & 4 \\ 5 & 6 & 7 \end{bmatrix}$$

Answer: D^{t} as shown above (order 2 × 3)

5. Find the negative of the following matrices:

Negative of a matrix: multiply every element by −1.

(i)

$$A = \begin{bmatrix} -3 & 0 \\ 5 & 6 \end{bmatrix}$$

$$-A = \begin{bmatrix} 3 & 0 \\ -5 & -6 \end{bmatrix}$$

Answer: −A = 3, 0/−5, −6

(ii)

$$B = \begin{bmatrix} -3 & 3 \\ -2 & 2 \end{bmatrix}$$

$$-B = \begin{bmatrix} 3 & -3 \\ 2 & -2 \end{bmatrix}$$

Answer: −B = 3, −3/2, −2

(iii)

$$C = \begin{bmatrix} -9 & 1 \\ 1 & -7 \end{bmatrix}$$

$$-C = \begin{bmatrix} 9 & -1 \\ -1 & 7 \end{bmatrix}$$

Answer: −C = 9, −1/−1, 7

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6. If A = 2, 3/4, 5 and B = 7, 6/5, 8, then verify that (i) $(A^t)^t = A$ (ii) $(B^t)^t = B$

(i) $(A^t)^t = A$

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$$

A^{t} =

$$\begin{bmatrix} 2 & 4 \\ 3 & 5 \end{bmatrix}$$

(A^{t})^{t} =

$$\begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$$

Answer: $(A^t)^t = A$, hence verified

(ii) $(B^t)^t = B$

$$B = \begin{bmatrix} 7 & 6 \\ 5 & 8 \end{bmatrix}$$

B^{t} =

$$\begin{bmatrix} 7 & 5 \\ 6 & 8 \end{bmatrix}$$

(B^{t})^{t} =

$$\begin{bmatrix} 7 & 6 \\ 5 & 8 \end{bmatrix}$$

Answer: $(B^t)^t = B$, hence verified

7. Show that L = 2, 3, 4/3, −2, 5|4, 5, 0 is a symmetric matrix.

A square matrix is symmetric when $A^t = A$, i.e. the element of row i and column j equals the element of row j and column i.

$$L = \begin{bmatrix} 2 & 3 & 4 \\ 3 & -2 & 5 \\ 4 & 5 & 0 \end{bmatrix}$$

L^{t} =

$$\begin{bmatrix} 2 & 3 & 4 \\ 3 & -2 & 5 \\ 4 & 5 & 0 \end{bmatrix}$$

Here $l_{12} = 3$

$= l_{21}$

$l_{13} = 4$

$= l_{31}$

$l_{23} = 5$

$= l_{32}$

Also the diagonal elements 2, −2, 0 stay in their places

Answer: $L^t = L$, so L is a symmetric matrix

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